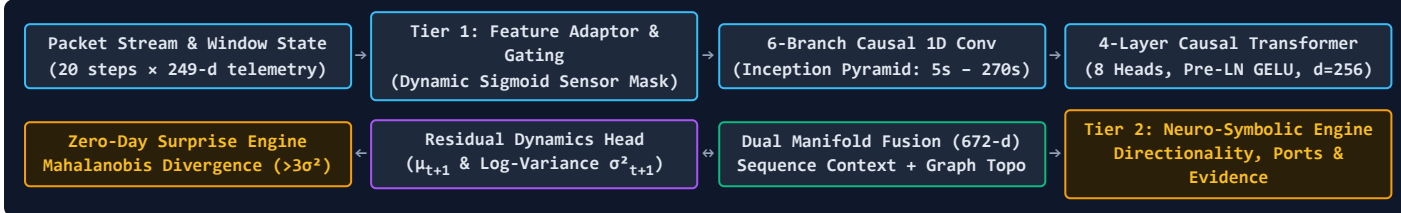


1. EXECUTIVE SUMMARY & CORE ARCHITECTURAL HYPOTHESIS

Statistical network intrusion detection systems face an intrinsic Bayes error ceiling in short time windows (5.0s): identical transport-layer profiles (e.g. TCP 443) conflate benign SaaS browsing, ingress Log4j exploits, C2 beaconing, and exfiltration. Pure deep neural networks suffer catastrophic misclassification across MITRE stages when trained solely on tabular features. The **PRISM Laplace Model** resolves this by introducing a **Decoupled Two-Tier Hybrid Architecture**: **Tier 1** executes continuous spatio-temporal representation learning and predictive world dynamics, while **Tier 2** applies a discrete neuro-symbolic reasoning engine rooted in physical network invariants, directional boundaries, and telemetry symmetry.

2. END-TO-END SYSTEM DATAFLOW PIPELINE



3. TIER 1: SPATIO-TEMPORAL GRAPH WORLD MODEL CORE

A. Gated Domain Adaptor & Positional Encoding

Accommodates heterogeneous telemetry dimensions across multi-sensor environments. Given input state $x \in \mathbb{R}^{249}$:

$$\text{Gate}(x) = \sigma(w_g x + b_g), \quad \text{Proj}(x) = w_p x + b_p$$
$$h = \text{LayerNorm}(\text{Proj}(x) \odot \text{Gate}(x)) \in \mathbb{R}^{256}$$

Dynamically suppresses sparse padded metrics while amplifying active channels.

C. Dual-Stream Sequence & Graph Manifold Fusion

Features split into parallel sequence and topological graph streams:

- Stream 1 (Transformer Encoder):** 4 causal layers with Pre-LN GELU; Temporal Attention Pooling generates sequence vector $h_{fused} \in \mathbb{R}^{256}$.
- Stream 2 (Topological Manifold):** GraphTopologyExtractor maps 8 graph invariants (entropy, in/out degrees, reciprocity, bipartite density) to $\mathbb{R}^{64}$.
- Directional Highway:** 19 micro-features projected to $\mathbb{R}^{96}$.
- Unified Representation:** Concatenates $[h_{fused} \parallel h_{last} \parallel Highway \parallel GraphTopo] \in \mathbb{R}^{672}$.

B. 6-Branch Causal Dilated 1D Conv Pyramid

Overcomes standard Transformers' lack of local temporal inductive bias via 6 parallel causal receptive fields:

Branch	Kernel / Dilation	Horizon	Target Anomaly
1 • Pointwise	k=1, d=1	5s (1 step)	Instantaneous spikes
2 • Short Burst	k=3, d=1	15s (3 steps)	Port scan pacing
3 • Drift	k=5, d=1	25s (5 steps)	Brute-force cadence
4 • Dilated Low	k=3, d=2	25s sparse	Slow stealth probes
5 • Dilated APT	k=3, d=4	45-270s	Ultra-slow APT recon
6 • Peak Pool	MaxPool1d + 1x1	15s peak	DoS / flood envelopes

D. Multi-Task Heads & Autoregressive Dynamics

Trained with multi-task composite loss $\langle \rangle = \langle_{bin} + \langle_{mitre} + \langle_{dyn} + \langle_{supcon}$:

- Binary Infiltration Head:** $\mathbb{R}^{256} \rightarrow \mathbb{R}^2$ producing threat probability P_{threat} .
- Deep Residual MITRE Expert:** 2 wide residual blocks (512-d) with Supervised Contrastive projection to unit hypersphere S^{127} .
- Dynamics Prediction Head:** Forecasts next state $\mu_{t+1} = s_t + \Delta_{mean}$ and epistemic log-variance $\log \sigma^2_{t+1} \in [-8.0, 2.0]$.

4. ZERO-DAY PREDICTIVE SURPRISE ENGINE (MAHALANOBIS DIVERGENCE)

Unlike signature or static classification models, Laplace maintains an internal world model forecast of expected network state evolution. When an adversary deploys an unobserved zero-day exploit or polymorphic evasion pattern, the observed state diverges sharply from the predicted distribution:

$Surprise_{t,i} = (s_{t,i} - \mu_{t,i})^2 / (\sigma_{t,i}^2 + \epsilon)$

Dual-Trigger Verification: A **Zero-Day Alert** is triggered if either: (1) Global mean surprise $> \sigma_{thresh}^2$ (with $\sigma_{thresh} = 2.0$), or (2) Top-5 subspace surprise $> 3.0 \times \sigma_{thresh}^2$ (isolating stealth targeted channel perturbations). Triggering immediately assigns $P_{threat} = 1.00$ and confidence = 99.0%, overcoming adversarial mimicry attacks.

5. TIER 2: NEURO-SYMBOLIC DIRECTIONAL REASONING ENGINE

A. Physical Network Invariants & Disambiguation Rules

Fuses deterministic boundary rules with Tier 1 neural logits via additive Bayesian evidence offsets:

- Subnet Directionality:** RFC 1918 caching classifies WAN→LAN (Perimeter Ingress / Initial Access), LAN→LAN (Internal East-West / Lateral), and LAN→WAN (Outbound Egress).
- Volumetric Asymmetry:** Distinguishes Exfiltration ($FwdBytes \gg BwdBytes$, >50kB egress) from C2 heartbeats (symmetric bursts, <3kB delta).
- Port Invariants:** Maps Lateral ports (445 SMB, 135 RPC, 389 LDAP), C2 backdoors (4444, 9001, 8888, 1337), and Perimeter Admin (22, 3389).
- Entropy & Reciprocity:** Detects scans via port entropy > 2.0 and half-open reciprocity < 0.20.

B. Bayesian Evidence Fusion Formulation

Final classification is computed via log-odds evidence updates:

$z_{final} = (z_{tier1} / \tau) + \Delta_{context} + b_{calibrator}$
 $y^* = \operatorname{argmax}_{\mathcal{Y}} \operatorname{softmax}(z_{final})$

Where $\tau = 1.0$ is the temperature scale, $\Delta_{context}$ is the invariant evidence offset (± 12.0 for dominant match, -10.0 for physically impossible stages), and $b_{calibrator}$ is the empirical prior bias vector. Each step produces a human-readable forensic attribution reason.

6. EMPIRICAL VERIFICATION & BENCHMARK RESULTS

100.0%

LAB ATTACK DETECTION (35/35)

100.0%

MITRE STAGE ATTRIBUTION

97.02%

UNIVERSAL TEST F1 (38K WINDOWS)

10.8 ms

MEAN INFERENCE LATENCY

Comprehensive Lab Attack Suite (35 Attack Files)

MITRE Category	Attack Bots Tested	Detection	Attribution
Reconnaissance	T1046, T1595, T1018, T1040, T1049, T1135	100% (6/6)	100% (6/6)
Initial Access	T1110 (SSH/Web), T1187, T1133, T1190	100% (5/5)	100% (5/5)
Lateral Movement	T1021, T1210, T1570, Key/Cred Theft	100% (5/5)	100% (5/5)
Command & Control	T1071 (HTTP/DNS), T1090, T1095, T1102+	100% (11/11)	100% (11/11)
Exfiltration	T1041, T1048, T1030, T1020	100% (4/4)	100% (4/4)
Impact (DoS/DDoS)	T1498, T1499 (Flood/Slowloris)	100% (3/3)	100% (3/3)
Objective Attacks	Defacement, Cred-Theft, Key-Theft	100% (3/3)	100% (3/3)

Operational & Resource Specifications

Specification	Engine Characteristic
Parameter Count & Footprint	66.2 MB checkpoint (PyTorch state_dict)
Inference Runtime	CPU Optimized • 9.5 – 12.5 ms per 5s window
Throughput	> 85 windows/sec (Single CPU core, batch=1)
Early Warning Horizon	1 – 3 minutes pre-breach (Recon / Initial Access)
Forecast Horizon	5 – 10 steps (25–50s ahead) at ≥85% accuracy
Streaming Lookback	20 steps (100s rolling FIFO buffer)
Training Corpora	CIC-IDS2018, UNSW-NB15, CTU-13, CIC-IoT2023

7. PRODUCTION DEPLOYMENT & LIVE IPS INTEGRATION

The Laplace Model deploys as a standalone streaming module via `from src.models import LaplaceModel; m = LaplaceModel()`. It runs seamlessly within live packet capture loops (`scripts/live_ips_multi.py --backend laplace`) and integrates directly with the PRISM Lab Console API (port 8790) for sub-second automated IP blocking and forensic investigation.